

THERE IS NO LARGEST PRIME NUMBERS

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OUTLINES

① PRIME NUMBERS

WHAT ARE PRIME NUMBER ?

DEFINITION

A prime number is a number that has exactly two divisors

EXAMPLE

- 2 is prime (two divisors: 1 and 2)
- 3 is prime (two divisors: 1 and 3)
- 4 is not prime (three divisors: 1, 2 and 4)

THERE IS NO LARGEST PRIME NUMBERS

THE PROOF USES *REDICTIO AD ABSURDUM*.

THEOREM

There is no largest prime numbers

PROOF.

- 1 Suppose p is the largest prime number.
- 2 Let q the product of the first p numbers.
- 3 Then $q + 1$ is not divisile be any of them.
- 4 But $q + 1$ is grreater than 1, thus divisible by some prime numbers not in the the first p prime number. □

WHAT STILL TO DO ?

ANSWERED QUESTION

How many prime are there ?

OPEN QUESTION

Is every even number the sum of two prime numbers ?

AN ALGORITHM FOR FINDING PRIME NUMBERS.

```
int main (void)
{
    std::vector<bool> is_prime (100, true);
    for (int i = 2; i < 100; i++)
        if (is_prime[i])
        {
            std::cout << i << " ";
            for (int j = i; j < 100; is_prime [j] = false, j+=i);
        }
    return 0;
}
```

Note the use of **std::** .